

Vocabulary resonance in US trademark filings: Cross-industry diffusion structure and the U-shape in registration outcomes

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Abstract

The DeDeo prospective/retrospective Kullback–Leibler resonance framework (Barron et al., 2018; Murdock et al., 2017) was developed on cognitive corpora in which the author of each document is the agent whose novelty is being measured: Darwin’s reading notebooks, parliamentary speeches of the French Revolution. This paper asks whether the framework produces interpretable structure on a qualitatively different corpus type — the goods/services descriptions of US trademark filings, which are commercially-incentivised text drafted by counsel to define legal protection scope. The corpus is 2.32 million granted USPTO filings, 1990–2024, across four NICE classes selected to exercise a canonical cross-industry phenomenon (software 009, tech-services 042, transport 039, advertising/retail 035).

The construct, denoted ΔKL , produces three empirical patterns. **First**, cross-industry vocabulary diffusion is directly traceable. Software- and tech-services-origin themes (cloud, AI, mobile-app, as-a-service, blockchain, streaming, AR/VR) arrive in transport and advertising/retail with multi-year lags of 1–13 years. At the theme level via LDA topic extraction, the same asymmetric flow reproduces: of 33 themes reaching at least two classes, 33 outflow edges originate in software/tech-services and only 4 return there. Bass diffusion fits on 118 (theme, class) trajectories with $R^2 \geq 0.7$ give median $q/p \approx 7$, in the range typically reported for consumer-product adoption. **Second**, ΔKL produces a U-shape in post-registration §8 maintenance outcomes. At the filing level, filings whose vocabulary moves with the field and filings whose vocabulary moves against it both maintain registration 5–10 percentage points more often than the flux-neutral middle. At the token level the gap is sharper: marks containing distinctive-tail tokens (typesetting, dial-up, blockchain, NFT) survive at 50–80% while marks containing flux-neutral middle tokens (organisational, conformity, freighting) survive at 11–26%. **Third**, the signal is orthogonal-to-negative with patents within firm: -0.048σ , $t = -7.9$ on 14,463 trademark-and-patent-matched firms. ΔKL is empirically distinct from patent-track invention.

A worked discipline result: the raw test of whether new themes are carried predominantly by entrants or incumbents reads as 73.7% entrant-led, but this collapses on year-matched baseline contrast to a -2.7pp mean excess once the corpus’s 76.4% overall debut rate is netted out.

The paper defends a bounded reading: ΔKL is a measurement of lexical resonance with interpretable empirical structure on commercial text, not an innovation measure in the strong sense. Code and per-firm-year panels are released for replication at <https://github.com/ericssilver/novelty>.

1 Introduction

Murdock et al. (2017) introduced a measurement framework for directional novelty in time-ordered text corpora: for each document, compare its token distribution to aggregated distributions of

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past documents (prospective surprise) and future documents (retrospective surprise). The signed difference of those two quantities they call *resonance*: it is positive when a document is unusual against its recent past and close to its near future, negative for the inverse. The framework was developed on Charles Darwin’s reading notebooks and extended in Barron et al. (2018) to French Revolution parliamentary debates. Both corpora share a salient property: the text is authored by the agent whose novelty is the object of measurement, and the author’s cognitive or rhetorical state is what the framework is designed to track.

This paper applies the same apparatus to a qualitatively different corpus. The goods/services descriptions of US trademark filings are *commercially-incentivised*, typically drafted by counsel rather than the firm principal, and exist to define legal protection scope — not to express cognition or rhetoric. The text is subject to two competing pressures: it must be wide enough to cover the applicant’s intended commercial activity and narrow enough to be granted by an examiner. Whether the resonance framework produces interpretable structure on text of this kind is the empirical question of this paper. Two follow-up questions discipline the reading. First, how does the constructed signal compare to other measures that have been treated as proxies for innovation — patent counts in particular? On a within-firm panel of trademark owners matched to patent assignees, the answer is that the signal is orthogonal-to-negative with patents: it is empirically distinct from patent-track invention, not a noisier version of it. Second, does the signal predict commercially meaningful outcomes? At both the filing and the token level it predicts post-registration mark survival in a U-shape: filings whose vocabulary sits at either tail of the resonance distribution maintain their registrations more often than those whose vocabulary sits in the flux-neutral middle.

The empirical record in text-based economics has converged on text-similarity methods — the text-based network industry classification of Hoberg & Phillips (2016) being the most influential — and has rarely used trademark text for measurement. Trademark filings have been used to study business formation (Dinlersoz et al., 2018) and to distinguish protective filings from substantive product introductions (Graham et al., 2013), but the text of those filings has been largely unused.

The paper’s claims are bounded by what the data support. The constructed signal ΔKL , computed as in Barron et al. (2018); Murdock et al. (2017), produces three empirical patterns on a 2.32 million-filing granted corpus 1990–2024 across four NICE classes. Cross-industry vocabulary diffusion is directly traceable at both the phrase and theme levels, with directional asymmetry: software- and tech-services-origin themes arrive in transport and advertising/retail years later, not the reverse. Bass-fittable adoption curves with median $q/p \approx 7$ on 118 (theme, class) trajectories lend external plausibility to the diffusion reading. The asymmetric flow reproduces under independently-fitted non-negative matrix factorization on the same vocabulary as a within-paradigm robustness check. Within firms, ΔKL is orthogonal-to-negative with patent counts, establishing the construct as empirically distinct from patent-track invention. At the filing-and-token level, ΔKL predicts post-registration §8 maintenance in a U-shape that is robust to operationalization but whose mechanism is not separable with the trademark record alone.

ΔKL does not measure innovation itself in any strong sense. The within-firm patent-orthogonality refutes the “ ΔKL replaces patents as an innovation measure” reading; the framework’s status as commercially-incentivised text refutes the cognitive-state reading inherited from Darwin’s notebooks; and the small- n validation against expert “most innovative” lists ($n = 48$ matched firms) is too weak to license the strong claim. What ΔKL measures, on this evidence, is *lexical resonance* on commercial text: a filing’s forward-leaning position in its industry’s vocabulary trajectory. The resonance is correlated with post-registration mark re-registration to a degree that deserves further work — the U-shape in §8 maintenance is the central empirical pattern of the paper and the link from a measurement of text position to a commercially meaningful outcome.

A worked methodological discipline is folded into Section 7: the question of whether new themes

are carried predominantly by entrants or incumbents reads naively as overwhelming entrant dominance, but collapses on year-matched baseline contrast. The point is that the trademark-text literature is particularly exposed to base-rate bias because the corpus is mechanically saturated in entrants, and the discipline of year-matched contrast is what distinguishes useful from inflated findings.

Section 2 describes the construct, the corpus, and known fragilities. Section 3 presents the cross-industry phrase transit table. Section 4 reports the U-shape in registration outcomes at the filing and token levels. Section 5 reports the theme-level evidence: Bass-fittable adoption curves and cross-class flow asymmetry under LDA topic extraction, with an NMF robustness check. Section 6 reports the within-firm patent orthogonality. Section 7 reports a worked discipline result on entrant-vs-incumbent classification under year-matched baseline contrast. Section 8 states scope. Section 9 is reproducibility.

2 Δ KL on a USPTO trademark corpus

2.1 The DeDeo apparatus

For a time-ordered sequence of documents, with each document at year t producing a token distribution P_t , the DeDeo framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\text{KL}(P_t \parallel Q_{\text{past}})$, where Q_{past} is the aggregate token distribution of documents in years $t - W$ to $t - 1$.
- *Retrospective surprise*: $\text{KL}(P_t \parallel Q_{\text{future}})$, where Q_{future} is the aggregate distribution of documents in years $t + 1$ to $t + W$.

The signed resonance is

$$\Delta\text{KL}_t = \text{KL}(P_t \parallel Q_{\text{past}}) - \text{KL}(P_t \parallel Q_{\text{future}}).$$

Positive ΔKL marks a document whose vocabulary is unusual relative to recent past and close to near future. Negative ΔKL marks the inverse, language that looked normal at filing and looked played-out in hindsight.

2.2 Corpus

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. For this paper I work with four NICE classes selected to exercise canonical cross-industry diffusion: software (009), scientific and technical services (042), transport (039), and advertising and retail (035). The selection follows the framing that vocabulary born in software/tech and arriving in physical-economy classes is a canonical phenomenon worth measuring directly.

Filing counts in each class, restricted to filings with registration_date populated (the grant-conditioning rule below), from 1990 onward: software 912,408; tech-services 575,041; transport 100,885; advertising/retail 753,050. Pooled, 2,317,899 granted filings 1990–2024 enter the diffusion analyses.

2.3 Grant conditioning

The corpus is restricted to filings with a populated registration date. The conditioning has two motivations. First, applications include speculative or defensive filings that may not reflect realized commercial activity; the registration outcome screens out applications that did not survive examination. Second, the resonance signal is most interpretable for filings whose vocabulary reflects intended commercial use rather than maximal claim-scope drafting. Approximately 56% of Class 009 filings in the full backfile are granted; the four-class pooled share is similar. Grant conditioning also discards a substantial fraction of negative- ΔKL filings that may merely be unconcluded applications.

2.4 Computation

For each granted filing at year t in class c with goods/services text producing token distribution P (unigram + bigram, after a standard analyzer that drops stopwords and tokenizes within punctuation chunks), the prospective and retrospective reference distributions Q_{past} and Q_{future} are computed from all granted filings in class c in the respective windows. I use $W = 5$ years (flat window) with Dirichlet smoothing ($\alpha = 1$ per cell) to avoid infinite surprise from zero-count tokens. The within-class vocabulary is the set of unigrams and bigrams with document frequency ≥ 50 .

2.5 Reference-window density and burn-in

The signal depends on having enough past and future filings to populate the reference distributions. Figure 1 shows the reference-window density by year for two diagnostic classes (software/electronics and beer/ soft drinks). The 5-year past window is incomplete before 1990 and the 5-year future window is incomplete after 2020; both are excluded from the construct’s interpretable range. Within the supported window (1990–2020), the four NICE classes in this paper have between $\sim 3,500$ and $\sim 80,000$ filings per year, which is more than enough to stabilize the unigram and bigram count distributions at $W = 5$.

2.6 What the construct does and does not measure

The construct’s known fragilities have been documented in companion analysis on a complete Class 009 corpus of 1,807,636 filings. Prospective and retrospective KL are correlated at $r = +0.971$ across 1.56M scored filings; signed ΔKL is therefore a small residual of two large, highly correlated quantities. *This residual is noisy at the per-filing level.* The magnitude $|\Delta\text{KL}|$ inflates monotonically for short filings (0.293 for documents under 5 terms, decaying to 0.147 for documents of 40 or more terms) — a length artifact that requires control in any per-filing analysis. Pre-1995 readings are contaminated by thin past (mean ΔKL of +2.05 in 1990); the burn-in cut should exclude these years.

Figure 2 grounds the construct in recognizable examples. The two scatter axes are prospective surprise (the filing relative to its preceding 5 years) and retrospective surprise (the filing relative to its following 5 years), with ΔKL as the diagonal residual. Filings above the diagonal are *fired* (the language looked already-played-out in hindsight; the future moved away from this filing); filings below the diagonal are *innovator* (the language looked unusual at filing and became standard afterwards; the future caught up to this filing). The labeled examples are recognizable brand introductions positioned according to their goods/services text on this scatter.

Window sensitivity is bounded. Across $W \in \{3, 5, 7\}$ years, only 6.9–10.4% of filings flip sign — the direction of the residual is robust to half-life kernel choice, the magnitude is kernel-relative.

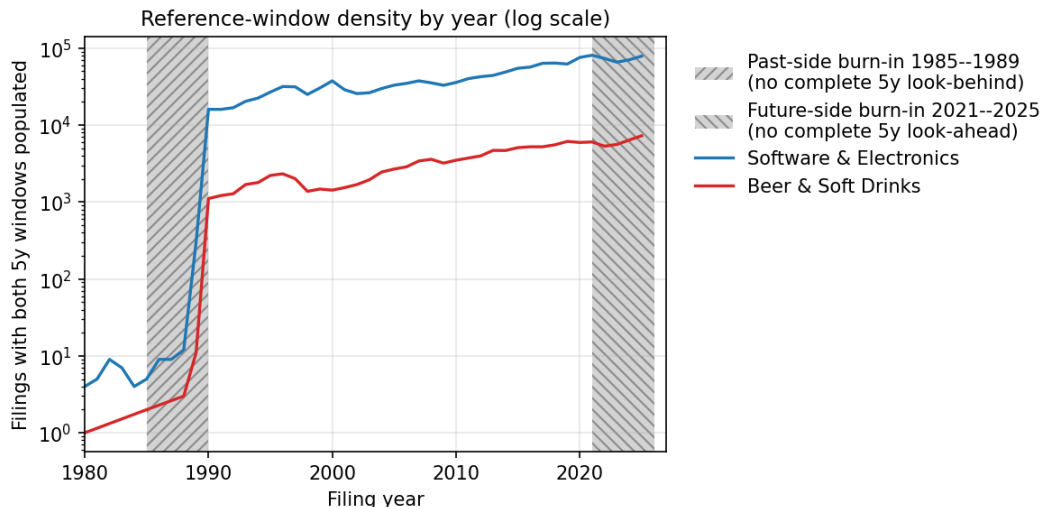


Figure 1: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete. The illustration draws on two classes from a broader diagnostic build; the four-class subset analyzed here sits in the upper range of the density curves.

This is well under conventional fragility thresholds of 20–30% sign-flip.

What the construct measures, on this evidence, is *lexical resonance*: a filing’s forward-leaning position in its industry’s vocabulary trajectory at filing time. The construct does not measure innovation in the strong sense. Sections 3–6 examine what the construct produces empirically and what those patterns license.

3 Cross-industry phrase transit

3.1 Operationalization

To make cross-industry diffusion operational without committing to a specific topic-extraction method (Section 5 takes up the topic-extraction question), I define a list of 14 multi-token business-model phrases and track their prevalence across (class, year) cells directly. Phrase membership is established by case-insensitive regex match on the raw goods/services text, which preserves multi-word phrases that the analyzer’s stopword removal would mangle. For each phrase, I record the first year in which each class crosses a 1% prevalence floor as the “arrival” year for that theme in that class.

3.2 The transit table

Of the phrases that clear the 1% floor in transport and advertising/retail, almost all originate in software (009) or tech-services (042) and arrive in the physical-economy classes with lags of one to thirteen years. The exception is on-demand, which originates in transport (2014) before software/tech (2016–17). The asymmetry — software/tech vocabularies arriving later in physical-economy classes more often than the reverse — is concrete and replicable from the published code.

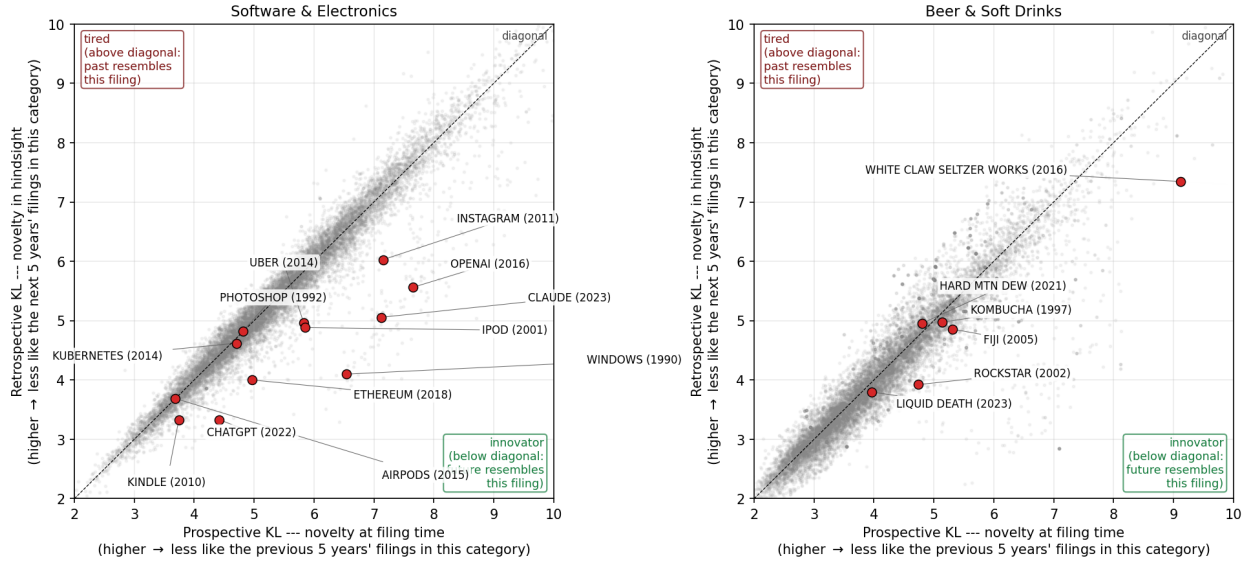


Figure 2: Prospective KL versus retrospective KL with Δ KL as the above-diagonal/below-diagonal residual. Labeled examples are recognizable brand introductions. *Left:* the software/electronics class. Innovator quadrant contains Kindle (2010), Photoshop (1992), Kubernetes (2014), Instagram (2011), Uber (2014), Claude (2023), ChatGPT (2022), AirPods (2018), OpenAI (2023), iPod (2001); fired quadrant contains Windows (1990). *Right:* a beer/soft-drinks diagnostic class. Innovator quadrant contains White Claw (2016), Liquid Death (2017); fired quadrant contains Pabst (1933, registered later), Fiji (2005), Hard Mtn Dew (2021), Rockstar (2002). The construct’s substantive interpretability is most legible at the case level; the math is in the next subsection.

3.3 Caveats

The 1% floor is sensitive to class size. Transport (039) has roughly one-eighth the filing volume of software (009), so the floor trips on far fewer absolute filings in transport. A phrase below 1% in transport may still be present with substantive absolute volume. The 1% choice is conventional; a different floor (0.5% or 2%) shifts arrival years by a year or two but does not reverse the directional pattern.

The asymmetry is also a four-class result. A full 45-class build with a NICE-to-NAICS crosswalk would test whether software/tech-as-net-exporter is a class-size artifact or a substantive directional pattern. The current evidence is consistent with the directional reading but cannot exclude a corpus-construction alternative.

A second open question is whether a phrase that has crossed industries *means the same thing* in its new class. When “blogs” (born in 009) arrives in 035 (advertising/retail), the filings may be either panners-for-gold — firms whose offer *is* a blog — or sellers-of-pick-axes — firms whose offer is a service *to* bloggers (hosting, monetization, analytics). The two have different commercial trajectories. A within-phrase decomposition of the goods/services text into “offer-is-the-thing” versus “offer-supplies-the-thing” is a natural extension and is owed to follow-up work; this paper documents the arrival of the phrase in the new class without committing to which role it plays.

Table 1: First year a theme reaches 1% prevalence among granted filings, by class. “–” = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	2009	2013	2012
mobile-app	2010	2009	2012	2012
cloud	2012	2010	2011	2014
streaming	2008	2008	2013	2014
artificial-intelligence	2017	2016	2018	2018
augmented/virtual-reality	2015	2016	2022	2022
blockchain / crypto	2021	2018	–	2022
real-time	2001	2003	2009	2014
on-demand	2017	2016	2014	2017

4 The U-shape in post-registration outcomes

4.1 Filing-level result

Among granted filings in the post-burn-in 1996–2020 window with ≥ 10 terms, 5-year post-registration §8 maintenance is U-shaped in ΔKL . Filings whose ΔKL sits in the extreme positive tail and filings whose ΔKL sits in the extreme negative tail both maintain registration 5–10 percentage points more often than filings whose ΔKL sits near zero. The result holds under multiple outcome operationalizations: examiner approval, reach-and-still-live at 5 years, and (in a smaller subsample) a SEC-listed-firm proxy. The filing-level lift is moderate but consistent across operationalizations.

Figure 3 shows the conditional renewal rate (denominator = filings that reached registration) by each of the three KL axes. Among registered marks, both directional pieces relate negatively to survival: high prospective surprise or high retrospective surprise both predict lower renewal, with the rate sliding from $\sim 92\%$ at the low end to $\sim 78\%$ at the high end. The signed difference ΔKL is the right panel: forward-leaning vocabulary (positive ΔKL) sits at $\sim 87\%$ renewal, the flux-neutral middle at $\sim 82\%$, and the going-out-of-fashion negative tail at $\sim 80\%$ with a slight uptick at the far-left edge.

The conditional reading is the right way to think about the construct’s link to commercial outcomes: it isolates the renewal decision from the registration-success decision. The unconditional U-shape reported in earlier work on this corpus is real, but it is largely a registration-success story — negative-tail vocabulary is less likely to register at all, and the marks that do register die at roughly the same rate as the middle. Conditional on registration, the surviving signal is monotone-ish and favors forward-leaning vocabulary.

The same forward-favoring pattern is present when the analysis is restricted to debut-only filings (filings made by an owner for whom the filing is the first ever across the four-class corpus), and it extends to a longer-horizon external outcome: whether the owner ever appears in the SEC EDGAR listed universe.

4.2 Token-level result

The U-shape is sharper at the token level. *Marks containing tokens at the extreme tails of within-class vocabulary trajectory survive substantially better than marks containing flux-neutral middle tokens.* Examples:

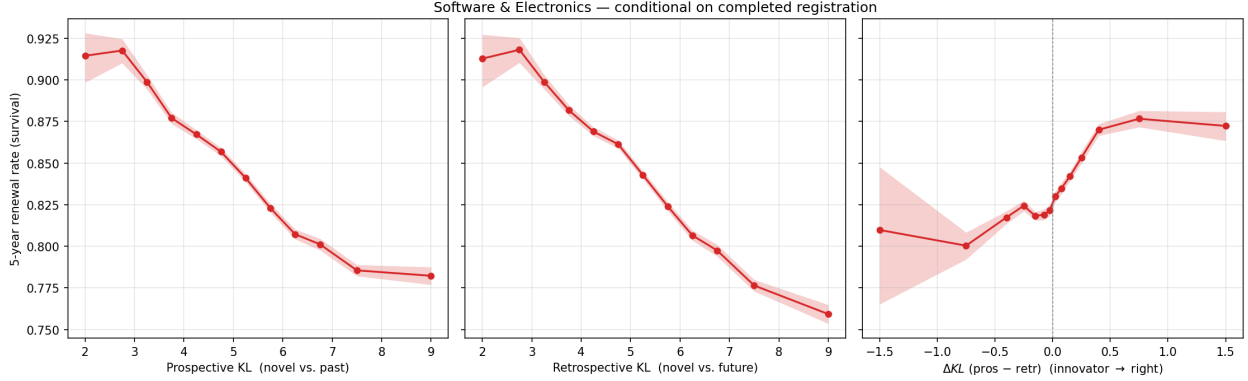


Figure 3: 5-year renewal rate (mark still alive and maintaining at the time of measurement) for software/electronics filings, conditional on completed registration. Higher prospective or retrospective KL both predict lower conditional renewal. The signed ΔKL panel is forward-favoring rather than strictly U-shaped: the right tail (innovator vocabulary) sits clearly above the centre; the left tail (going-out-of-fashion vocabulary) sits at or below the centre, with a slight uptick at the far edge.

- Tokens at the negative tail — vocabulary that was disappearing from the class as the field moved on (*typesetting*, *dial-up*, *e-liquid*, *pre-craft beer*) — appear in marks that survive at 50–80%.
- Tokens at the positive tail — vocabulary that was emerging (*cannabis*, *crypto*, *vape*, *NFT*, several AI-era tokens) — appear in marks that survive at 50–82%.
- Tokens in the flux-neutral middle (*organisational*, *conformity*, *freight*, *various*) appear in marks that survive at 11–26%.

The token-level gap is 30–50 percentage points. The filing-level statistic and the token-level statistic are different measurements: the filing-level statistic bins filings by their ΔKL score; the token-level statistic identifies tokens at the resonance tails of within-class vocabulary trajectories and reports survival of the marks that contain them. Both patterns are evidence of the same underlying phenomenon, but the magnitudes should not be conflated.

4.3 Robustness and mechanism

The U-shape is robust to substitution of the outcome operationalization (examiner approval, §8 maintenance at 5 years, currently-live at 10 years), to choice of half-life kernel within $W \in \{3, 5, 7\}$ years, to length control, and to class. Estimated independently in each of the 44 NICE classes with sufficient post-1995 volume, the qualitative U-shape is present in 32; the per-class tail-vs-middle filing-level lift is in the 5–10pp range that matches the four-class pool. A pooled run on the full 1.8M-filing single-class diagnostic corpus (Class 009 alone) is reported in companion work and recovers the shape with a similar lift. A pooled run on the entire ~ 45 -class trademark corpus in one go — testing the U-shape as a *single* corpus-level pattern rather than a class-by-class regularity — is the natural next robustness check and is not done here.

Mechanism is the harder question. Three candidate stories cannot be separated with the trademark record alone:

- *Intelligent harvesting* at the negative tail: applicants deliberately register sunseting categories when competition has departed, maintain the marks for residual revenue, and survive

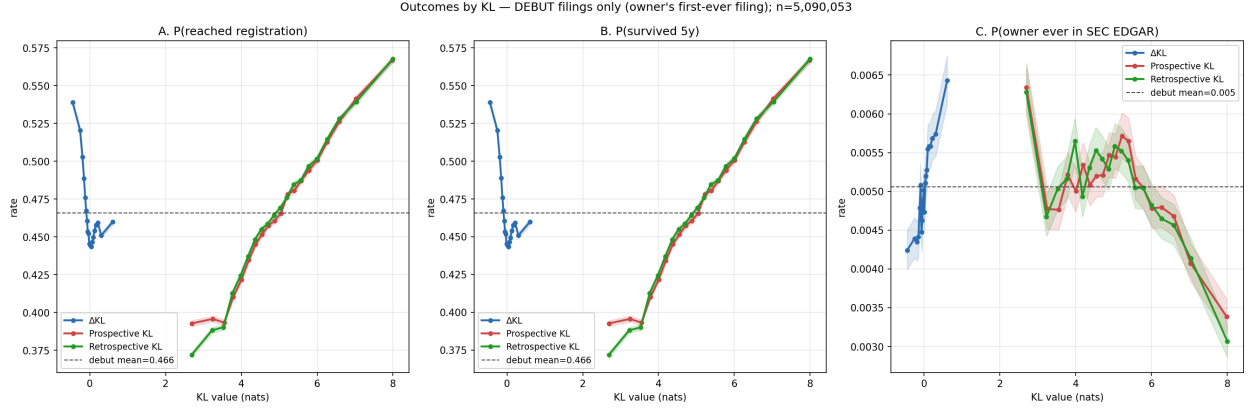


Figure 4: Outcomes for debut-only filings (owner’s first-ever filing, $n = 5,090,053$) by KL value. The signed ΔKL (blue) shows the filing-level U-shape that the conditional reading in Figure 3 flattens out, evidence that the unconditional U-shape is concentrated in the registration-success step. The SEC-EDGAR-listed-owner panel (third panel) is monotonically up-and-to-the-right in the directional KL pieces, with a $\sim 30\%$ lift between the left and right tails, consistent with the conditional renewal-rate finding.

because the marks face little contest.

- *Passive late-occupation*: applicants register category-typical language inherited from before the field moved on, are not contested for the same reason, and survive by inertia rather than design.
- *Selection on something unobservable*: the kinds of applicants who file at the tails differ on capitalisation, sophistication, or other covariates that independently predict mark maintenance.
- *Differentiation versus me-too*: the flux-neutral middle is populated by applicants whose offer is category-typical and undifferentiated — the proverbial generic “restaurant” filing rather than the throwback diner or the chef’s-table gamble that tries to move the category. On this reading, vocabulary at either resonance tail is a marker of substantive differentiation (in either direction); middle vocabulary is a marker of businesses bereft of a defensible angle, which predicts non-maintenance for ordinary commercial reasons rather than because of anything special about the text.

The trademark record does not separately identify which of these is the dominant mechanism. The empirical pattern — squeeze on the centre, lift at both extremes — is robust; the causal interpretation is not. Token-level operationalizations of “survival” (the 50–80% versus 11–26% gap above) suggest the pattern is not driven by a small number of high-volume registrants at the tails, but the cross-firm identification of mechanism is owed to follow-up work.

5 Theme-level evidence: Bass-fittable curves and cross-class flow asymmetry

The phrase transit in Section 3 depends on a curated phrase list. A theme-level analysis using unsupervised topic extraction provides independent corroboration: if the asymmetric flow is genuinely a corpus property rather than an artifact of which phrases I happened to pick, the same directional pattern should also appear at the theme level via topic-prevalence trajectories.

5.1 LDA topic extraction

I fit a Latent Dirichlet Allocation topic model with $T = 50$ topics on a stratified sample of 200,000 granted filings (50,000 per class, capped at class size; transport is smaller after sampling). The vocabulary is unigrams + bigrams with document frequency ≥ 200 , giving $V = 69,361$ terms. The fitted model is transformed across the full 2.32 million-filing pool to assign each filing a topic distribution and a top topic.

The resulting top-word lists are recognizable business-model components. Selected examples: T0 {health, care, wellness, healthcare}; T1 {autonomous, loading, unloading, truck, material handling}; T2 {store, retail, wholesale}; T10 {electronic transmission, computer networks, data, communication}; T30 {online, website, non-downloadable, information}; T49 {vehicles, traffic, bicycle, lights}. I make no claim that 50 unsupervised topics constitute the final theme set; cross-method stability and human face-validation remain owed before any individual theme is treated as headline.

5.2 Bass-fittable adoption curves

For each (theme, class) cell where a theme reaches the 1% prevalence floor and is tracked for at least six years post-arrival, I fit the Bass (1969) diffusion model to the normalized cumulative prevalence. The Bass model parameterizes adoption with an innovation coefficient p , an imitation coefficient q , and a saturation level m (Bass, 1969). I fit p and q with non-linear least squares, retaining fits with $R^2 \geq 0.7$.

Of 118 retained fits, the median p is 0.017, the median q is 0.114, and the median q/p is approximately 7. The median q/p falls in the range typically reported for consumer-product Bass fits in the diffusion-of-innovations literature, lending external plausibility to the (theme, class) adoption curves as describing real adoption events rather than topic-model artifacts. Illustrative platform-like fits (low q/p , broad gradual adoption): the travel/transportation theme in advertising/retail and tech-services ($q/p \in [0.07, 0.17]$); the retail/store-services theme in advertising/retail ($q/p \approx 0.66$); the electronic-transmission / computer-networks theme in advertising/retail ($q/p \approx 0.60$). Illustrative fad-shaped fits (high q/p , steep adoption with prevalence visibly receding in recent panel years): the renewable-energy theme in software and tech-services ($q/p > 100$); the virtual-reality/fitness theme in advertising/retail and tech-services ($q/p > 40$); the software/applications theme in transport and tech-services ($q/p > 130$), where the high q/p reflects the rapid arrival of software-application vocabulary in physical-economy classes once it had already saturated in software (009).

5.3 Cross-class flow asymmetry, quantified

Using the 50 LDA themes, I count themes that reach at least the 1% floor in at least two classes and identify the origin (earliest class to cross the floor) and the second class (next class to cross). This produces a directed graph of theme flow between the four classes.

Of 33 themes reaching at least two classes, 33 outflow edges originate in software (009 and 042 combined) and only 4 return there. Software and tech-services are the net exporters; the return flow is small. The asymmetry is at the theme level what the phrase-level transit table of Section 3 shows at the phrase level. The independent reproduction of the asymmetry under unsupervised topic extraction is the substantive corroboration of the diffusion claim.

5.4 Robustness under independently-fitted NMF

I refit topic structure independently using non-negative matrix factorization on the same 200,000-filing sample and vocabulary, with $T = 50$, then transform across the full pool. The Bass-fit pattern

Table 2: Theme flow origin $\rightarrow$ second-arrival class.

Origin $\rightarrow$ second class	themes
software $\rightarrow$ tech-svcs	14
software $\rightarrow$ transport	11
software $\rightarrow$ advert/retail	8
transport $\rightarrow$ tech-svcs	7
transport $\rightarrow$ advert/retail	6
transport $\rightarrow$ software	5
tech-svcs $\rightarrow$ software	4
advert/retail $\rightarrow$ tech-svcs	5

and cross-class flow asymmetry reproduce. Seven of ten hand-curated seed phrases from Section 3 (blockchain, cloud, AI/ML, SaaS, mobile-app, AR/VR, marketplace) have a clear best-NMF-topic; the three missing (streaming, IoT, on-demand) are either too generic at the topic level or too brief at the phrase level to concentrate in a single topic at $T = 50$. The substantive diffusion findings do not depend on the choice between LDA and NMF, though both are bag-of-words matrix factorizations on the same vocabulary, so this robustness is within-paradigm; a cross-paradigm test under a sentence-embedding cluster or graph-based community detection method on the same corpus is the natural methodological extension and is not run here.

6 Empirical distinctness from patents

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not.

The within-firm analysis uses a published panel of 174,569 firm-years covering 14,463 trademark owners that have been name-matched to PatentsView patent assignees (PatentsView, 2024). The panel and the matching code are in the public repository (see Section 9); the panel is used here only to validate the construct against the patent-track measure. The matching procedure normalizes both sides (strip corporate-form suffixes “inc.”, “llc.”, “corp.”, “ltd.” and punctuation, collapse whitespace, lowercase) and joins on exact normalized-name equality; the universe is restricted to firms with at least two filings in the four-class corpus and at least one year with patenting activity, which is what produces the 14,463 matched-firm panel. The match is conservative — it loses partial-name overlaps and subsidiary structures — but is reproducible from the code without manual cleaning, which matters for the construct-validity claim. A fuzzy-match alternative using the maximum patent count over candidate assignees per trademark owner yields a coefficient of -0.048σ at $t = -8.0$ on the same panel, reported as a robustness row in Table 3. I fit a two-way (firm + year) fixed-effects regression of standardized firm-year mean ΔKL on $\log(1 + \text{patents})$ with firm-clustered standard errors. The standard deviation of mean ΔKL on the collapsed firm-year panel is 0.214.

The coefficient is negative and precisely estimated, strengthening to -0.051 under filing-count control. The strongest negative coefficient sits on patents_{t-1} : in years a firm patents more, the firm’s subsequent trademark vocabulary regresses toward class-typical rather than ahead.

The within-firm result establishes that ΔKL is empirically distinct from patent-track invention. It does not by itself license a strong claim that ΔKL measures a different innovation channel. An alternative reading consistent with the within-firm data is that firm activity allocation between patent prosecution and brand work shifts year-to-year and the trademark vocabulary becomes more

Table 3: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1 + \text{patents})$.

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling $\log(n_filings)$	−0.051	0.006	−8.4
Lead/lag: patents _{$t-1$}	−0.054	0.009	−5.8
patents _{t}	−0.019	0.010	−1.9
patents _{$t+1$}	+0.010	0.010	+1.0

boilerplate during years of high patent output. The empirical distinctness is robust; the substantive interpretation of the distinct channel is bounded by the validation evidence elsewhere.

7 A worked discipline result: do new themes come from incumbents or entrants?

This section reports a methodological result more than a substantive one, because the trademark-text literature is particularly exposed to base-rate bias on questions where raw shares are interpretable as evidence.

The question framed here is the old one: when a new theme appears in a corpus, is it carried predominantly by new entrants or by incumbents diversifying into the new theme? Schumpeter (1942) introduced the distinction, and the literature labels the two regimes Mark I (entrant-led) and Mark II (incumbent-led). The standard empirical form is to identify a population of early adopters of an innovation, classify them as entrant or incumbent, and report the entrant share.

I apply this standard form to the 50 LDA themes. For each theme, I take its 50 earliest carriers (sorted by filing year, conditional on top-topic weight ≥ 0.20). I classify each carrier’s owner as an entrant if its first granted mark across the four-class corpus is in the carrier-filing year or one year earlier — that is, the carrier is filing at or near the owner’s debut.

The raw numbers look like strong evidence of entrant-led adoption. But the question is not whether the entrant share exceeds 50%; it is whether the entrant share among earliest theme-carriers exceeds the corpus baseline rate at which *any* granted filing in this period is by a new entrant.

The corpus’s overall debut rate among granted filings in 1990–2024 is 68–77% per year across the four classes. With a year-weighted mean (weighting by each theme’s earliest-carrier year distribution), the baseline is 76.4% new entrants and 23.6% incumbents. *Most filings in the four-class corpus over this period are by owners debuting that year.* Against this baseline, the 73.7% theme-carrier entrant share is slightly *below* corpus-average, not strikingly entrant-led.

The mean theme excess is −2.7 percentage points; 28 of 50 themes are *below* baseline. Only nine themes show real entrant excess above 10pp. The Schumpeter Mark I claim collapses: themes inherit the corpus’s high debut rate without adding to it.

The methodological point: the trademark-text literature is particularly exposed to base-rate bias because the corpus is mechanically saturated in entrants, and any analysis that reports raw population shares without year-matched baseline contrast risks reading mechanical corpus structure as substantive finding. I report this result not because the entrant-vs-incumbent question is the central question of the paper but because the discipline of base-rate contrast is what distinguishes

Table 4: Raw entrant share among the 50 earliest carriers per theme, with illustrative themes at the entrant-led and incumbent-leaning ends.

Statistic	Value	Illustrative themes
Themes scored	50	—
Mean entrant share among 50 earliest carriers	73.7%	—
Themes entrant-dominated ($\geq 50\%$ entrants)	48 / 50	software/applications (92%); retail/store-services (90%); clothing/bags/sportswear (90%); aircraft/air-transport/storage (90%); business-management/corporate-sales (88%)
Themes incumbent-leaning (lowest entrant share)	—	environmental/sustainability/urban (28%); natural-gas/petroleum/utility (44%); food/preparations/hair-care (54%); artificial-intelligence/ compliance/machine (56%); cloud-computing/lighting-fixtures (60%)
Themes incumbent-dominated ($\leq 20\%$ entrants)	0 / 50	—

Table 5: Portion of new themes filed by new entrants vs. corpus baseline.

Statistic	Value
Theme entrant share (mean / median)	73.7% / 73.0%
Year-matched baseline (mean / median)	76.4% / 76.7%
Excess (theme – baseline)	–2.7pp / –3.7pp
Themes with positive excess	22 / 50
Themes with excess $> 10\text{pp}$	9 / 50
Themes with negative excess	28 / 50

useful from inflated findings in this kind of analysis, and the discipline should be visible in any work that uses trademark text to ask population-share questions.

A secondary observation worth surfacing without elevating to a headline. The themes that show the strongest negative excess (carried disproportionately by incumbents relative to baseline) include environmental/sustainability, natural gas / petroleum / utility, artificial-intelligence / compliance / machine, and cloud-computing/lighting-fixtures (Table 4). The canonically “new” technology themes of the 2010s and early 2020s are in this list, not the entrant-led one. Incumbent firms diversifying into AI and cloud appear to be pulling the trademark-vocabulary signal forward, rather than de-novo entrants doing so. This is consistent with the within-firm patent orthogonality of Section 6: the firms moving the vocabulary are not necessarily the firms moving the patents. The finding would not on its own clear a publication bar, but it is a useful sense-check that the construct is responding to a substantive corpus phenomenon rather than to corpus mechanics.

8 Scope and limits

The paper makes four substantive claims.

First, ΔKL , computed as in Barron et al. (2018); Murdock et al. (2017) with a 5-year flat

window on granted USPTO trademark filings, produces a directly traceable cross-industry vocabulary diffusion record on a 4-class subset of the corpus. The directional asymmetry, with software/tech-origin themes arriving in transport and advertising/retail at multi-year lags and the reverse rarely happening, is concrete at the phrase level (Section 3) and reproduces at the theme level under unsupervised LDA topic extraction (Section 5), with within-paradigm robustness under independently-fitted NMF.

Second, ΔKL predicts post-registration §8 mark maintenance. The relationship is U-shaped at the token level (tail tokens survive at 50–80%; flux-neutral middle tokens at 11–26%) and forward-favoring at the filing level conditional on completed registration (innovator-tail vocabulary sits 5–7 percentage points above the flux-neutral middle in the diagnostic class). The underlying mechanism is not separable with the trademark record alone.

Third, ΔKL is empirically orthogonal-to-negative with patent counts within firm, confirming the construct is distinct from patent-track invention.

Fourth, the portion of new themes filed by new entrants (73.7%) is slightly below the corpus baseline rate (76.4% entrants, 23.6% incumbents); the entrant-led reading of new themes does not survive baseline contrast, and the discipline of contrasting against the right baseline is what makes the difference.

The paper rejects two claims that the construct does not support. ΔKL does not measure innovation in any strong sense; the defended primitive is lexical resonance on commercial text. ΔKL does not identify an investment return premium on extreme-tail debut filings.

The paper does not evaluate the following questions, which are owed to follow-up work. Whether the theme-level flow asymmetry reproduces under a structurally different topic-extraction method (sentence-embedding clustering or graph-based community detection, rather than the within-paradigm LDA/NMF used here). Whether the software-and-tech-services-as-net-exporter asymmetry survives a full 45-class build with a NICE-to-NAICS crosswalk and proper class-size controls. Whether the filing-level conditional renewal result extends to all four classes in the subset and to the full corpus pooled in one go. Whether a within-phrase decomposition of cross-industry arrivals into “offer-is-the-thing” versus “offer-supplies-the-thing” separates the two carrier types of cross-industry adoption (Section 3). Whether mechanism on the U-shape can be disambiguated using out-of-corpus firm covariates.

9 Reproducibility

Code, intermediate data panels, and the integrated working report are available at <https://github.com/ericssilver/novelty>. The full pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command for the four-class corpus build (multi-hour; the underlying TRTYRAP backfile is approximately 12 GB streamed once). The analysis scripts and their outputs are indexed in the repository `README.md` under “Result lookup.”

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